

Chapter 6: Vector Spaces, Span, Basis, Rank and Null Space

6.1 Introduction

In the previous chapter, we learned about scalars, vectors, matrices, and basic matrix operations. While these concepts help us represent data, modern AI algorithms require a deeper understanding of how vectors relate to one another.

Questions such as the following naturally arise:

- Can one vector be created from other vectors?
- How many independent features exist in a dataset?
- Are some features redundant?
- Why do some matrices have infinitely many solutions?
- How does PCA reduce dimensionality?

The answers to these questions lie in the concepts of **vector spaces, span, basis, rank, and null space**.

6.2 Vector Space

Definition

A **vector space** is a collection of vectors that satisfies two important properties:

1. Vector Addition

If

$$u, v \in V$$

then

$$u + v \in V$$

2. Scalar Multiplication

If

$$v \in V$$

and

$$c \in \mathbb{R}$$

then

$$cv \in V$$

Intuition

Imagine a large empty space.

Every possible direction you can move represents another vector.

As long as

- *adding vectors stays inside the space*
- *multiplying vectors by numbers stays inside the space*

the collection forms a vector space.

Example

Consider

$$\mathbb{R}^2$$

Every point

$$(x, y)$$

belongs to this vector space.

Examples

$$\begin{aligned} (2, 3) \\ (-, 15) \\ (100, -45) \end{aligned}$$

All belong to

$$\mathbb{R}^2$$

Interpretation

Every data sample is represented as a vector.

Example

Student

$$\begin{aligned} (\text{Age}, \text{Height}, \text{Weight}) \\ (20, 170, 65) \end{aligned}$$

belongs to

$$\mathbb{R}^3$$

6.3 Linear Combination

Almost every concept in Linear Algebra begins with a **linear combination**.

Suppose

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Then

$$3v_1 + 2v_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

The new vector is called a **linear combination**.

General equation

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n$$

6.4 Span

Definition

The **span** of a set of vectors is the collection of **all possible linear combinations** of those vectors.

Mathematically,

$$\text{Span}(v_1, v_2) = \{av_1 + bv_2\}$$

where

$$a, b \in \mathbb{R}$$

Example 1

Suppose

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Then

$$\text{Span}(v_1, v_2) = \mathbb{R}^2$$

These two vectors can generate every point in the plane.

Example 2

Suppose

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

Notice

$$v_2 = 2v_1$$

Therefore

they lie on the same line.

Their span is **only one line**, not the whole plane.

Interpretation

Suppose

Height

Weight

BMI

Since

BMI depends on Height and Weight,

adding BMI does not create a new direction.

The span remains unchanged.

6.5 Linear Independence

Vectors are linearly independent if

no vector can be written using the others.

Suppose

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Independent.

Now

$$v_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Notice

$$v_3 = v_1 + v_2$$

Hence

$$v_3$$

is dependent.

6.6 Basis

A **basis** is

- a set of linearly independent vectors
- that span the entire vector space.

For

$$\mathbb{R}^2$$

the standard basis is

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Why Basis Matters

The basis represents the **minimum number of vectors needed to describe every vector in the space.**

6.7 Dimension

The number of basis vectors is called the **dimension**.

Examples

$$\mathbb{R}^2$$

Dimension = 2

$$\mathbb{R}^3$$

Dimension = 3

6.8 Rank of a Matrix

The **rank** of a matrix is
the number of linearly independent rows or columns.

Example

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Notice

Row 2

=

2 × Row 1

Therefore

Rank

$$(A) = 1$$

Example

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Rows are independent.

Rank

$$(A) = 2$$

Interpretation

Rank tells us

how much useful information exists.

High rank



More information.

Low rank



More redundancy.

6.9 Column Space

The **column space** is the span of all columns.

Suppose

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

Columns

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

Since

Column 2

=

2 × Column 1

Column space is one-dimensional.

6.10 Row Space

The row space is the span of all rows.

Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$$

Only one independent row.

Hence

Row space dimension = 1.

6.11 Null Space

The **null space** is one of the most important concepts in Linear Algebra.

Definition

The null space contains every vector

$$x$$

such that

$$Ax = 0$$

Example

$$A = \begin{bmatrix} 1 & 2 \end{bmatrix}$$

Find

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

such that

$$x_1 + 2x_2 = 0$$

Choose

$$x_2 = 1$$

Then

$$x_1 = -2$$

Therefore

$$x = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

belongs to the null space.

Interpretation

These vectors disappear after multiplication.

They produce

$$0$$

Meaning

The null space represents

- redundant information
- hidden relationships
- degrees of freedom

6.12 Rank–Nullity Theorem

One of the most beautiful equations in Linear Algebra is

$$\text{Rank}(A) + \text{Nullity}(A) = \text{Number of Columns}$$

Example

Suppose

A

has

5 columns

Rank = 3

Then

Nullity

= 2